

SIMATS ENGINEERING

ASSESSMENT TOOL 1 : Concept Mapping

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE:CSA1502

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I KALPANA J (192521038) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student : *J. Kalpana*

Assessment Tasks Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Total Marks:

Permitted Tools :

draw.io / diagrams.net, Lucidchart, MindMeister, Miro, Canva, PowerPoint, Google Drawings, or equivalent diagramming tool

Procedure for Preparing the Digital Concept Map

1. Use the following procedure to prepare the diagram and upload evidence.
2. Students who already know another diagramming tool may use it, but the exported evidence and source file must still be submitted.
3. Open <https://app.diagrams.net/> or the desktop version of draw.io / diagrams.net.
4. Choose Device, Google Drive, or OneDrive as the save location according to availability.
5. Create a blank diagram and name the file as
CSA15_CO1_AT1_RegisterNumber_Concept_Map.drawio.
6. Place the concept map title at the center or as a clearly visible anchor node.

7. Add major cloud nodes: cloud definition, cloud evolution, cloud architecture, service models, deployment models, virtualization, web services, SOAP, REST, APIs, elasticity, scalability, resource pooling, measured service, and emerging paradigms.
8. Connect nodes with labeled arrows such as uses, depends on, enables, scales through, communicates through, belongs to, supports, provides, consumes, or controls.
9. Use colors meaningfully: one color for service models, one for cloud characteristics, one for API/web-service concepts, and one for concept map title-specific elements.
10. Export the final diagram as PNG or PDF. Keep the editable .drawio source file also.
11. Paste the exported diagram image or screenshot in this DOCX under Task 1.
12. Upload the DOCX, exported PNG/PDF, and .drawio file to Google Classroom.

Task 1: Concept Map

Create a concept map in draw.io, Lucidchart, Miro, Figma, or any suitable diagramming tool. The map must connect cloud ecosystem, cloud actors, service models, deployment models, virtualization, web services, REST/SOAP APIs, scalability, elasticity, and the student's Project.

Task 2A: Service Model Responsibility Matrix

Prepare a technical matrix that compares SaaS, PaaS, IaaS, FaaS/serverless, database-as-a-service, storage-as-a-service, identity-as-a-service, and monitoring/logging services. For each selected service category, explain what is managed by the cloud provider, what is managed by the student application team, and why that split fits the Project.

Task 2B: Cloud Service Decision Matrix

Map Project modules to cloud services. Include authentication, frontend hosting, backend/API layer, database, object storage, analytics/logging, notification, AI/Python module, security, CI/CD, backup, and monitoring. Students should justify the most relevant service choices instead of forcing every possible cloud service into the design.

Task 3: API Flow, Elasticity, and Scalability Design

Draw a professional cloud-flow diagram for the Project modules. The diagram must show user app or web app, API gateway/backend service, authentication, compute layer, database, object storage, analytics/logging, monitoring, security boundaries, and scaling points. Students may replicate the given visual pattern, but all components and labels must be customized to their own architecture.

Task 4: Technical Justification

Write a concise justification explaining how the diagram and matrices prove CO1. The justification must discuss service model selection, API role, elasticity, scalability, virtualization or managed compute choice, and cloud responsibility boundaries.

1. Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?
2. Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?
3. Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?
4. Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?
5. Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Expected Digital Evidence

- Editable diagram source file and exported PNG/PDF for the concept map.
- Completed service model responsibility matrix and cloud service decision matrix.
- Architecture/API flow diagram exported as PNG/PDF.
- README file containing title, tools used, repository structure, and a short CO1 reflection.

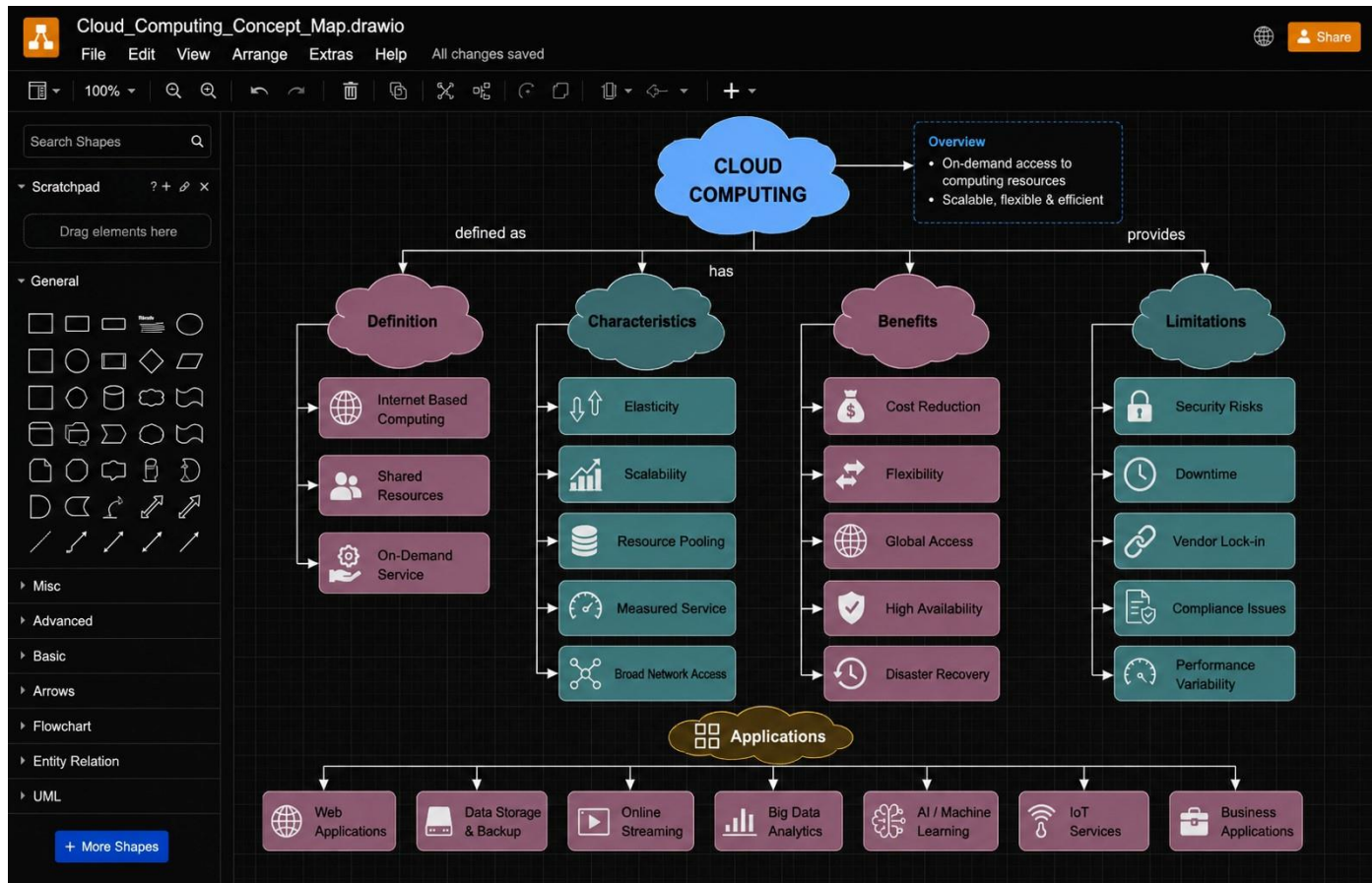
GitHub and Google Classroom Submission

1. Create a repository using a clear name: CSA15_CO1_AT<Number>_<RegisterNumber>.
2. Upload editable source files, exported PDF/PNG evidence, screenshots, and a short README explaining the work.
3. Keep filenames professional and include the register number where appropriate.
4. Submit only the GitHub repository link in Google Classroom before the deadline.
5. Ensure the repository is accessible to the faculty evaluator during assessment.

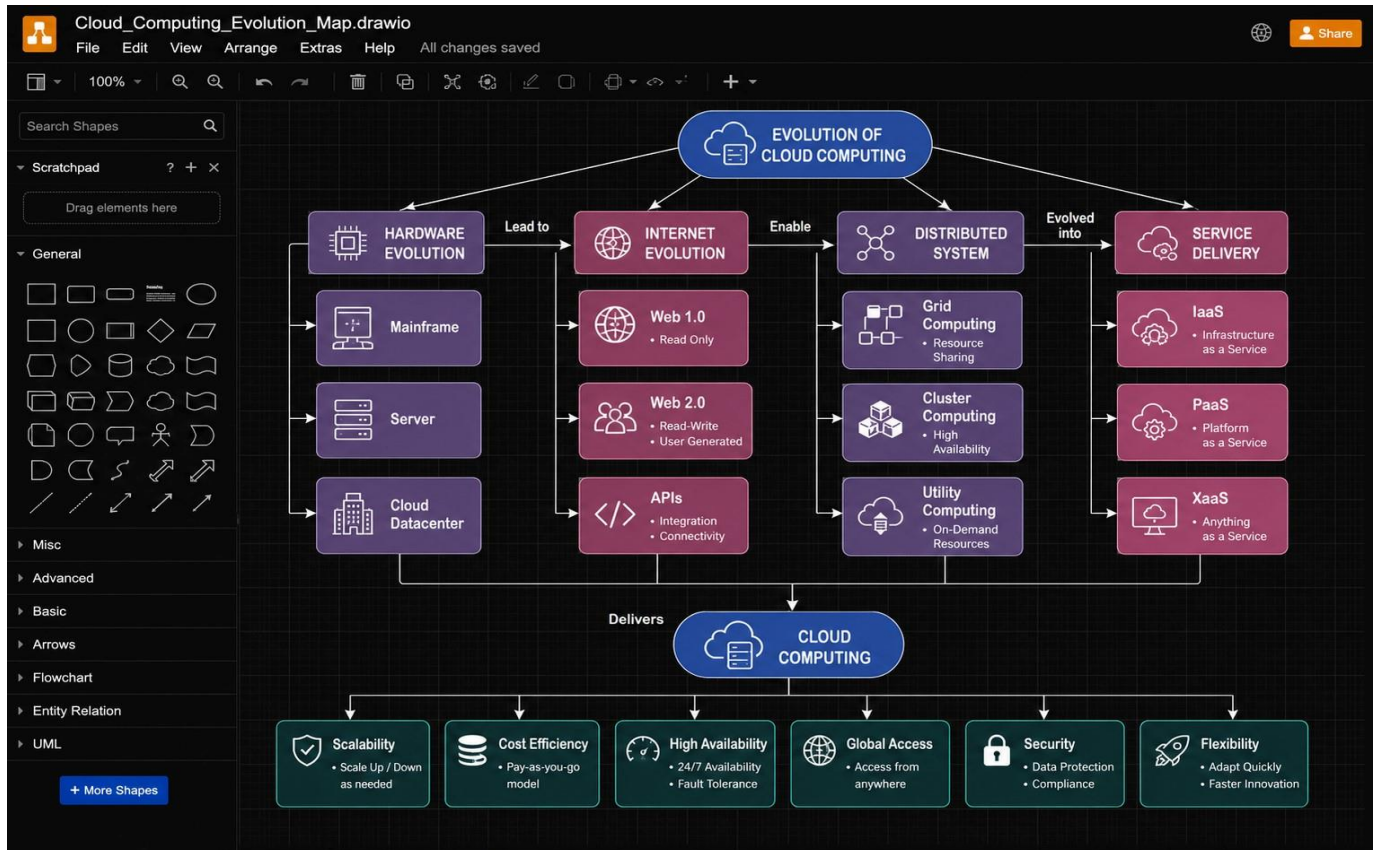
Implementation Note

Faculty may evaluate the submitted evidence digitally using repository inspection, diagram quality, oral explanation, and CO-aligned rubric scoring. The focus is on technical understanding, cloud design reasoning, and hands-on evidence rather than routine written work.

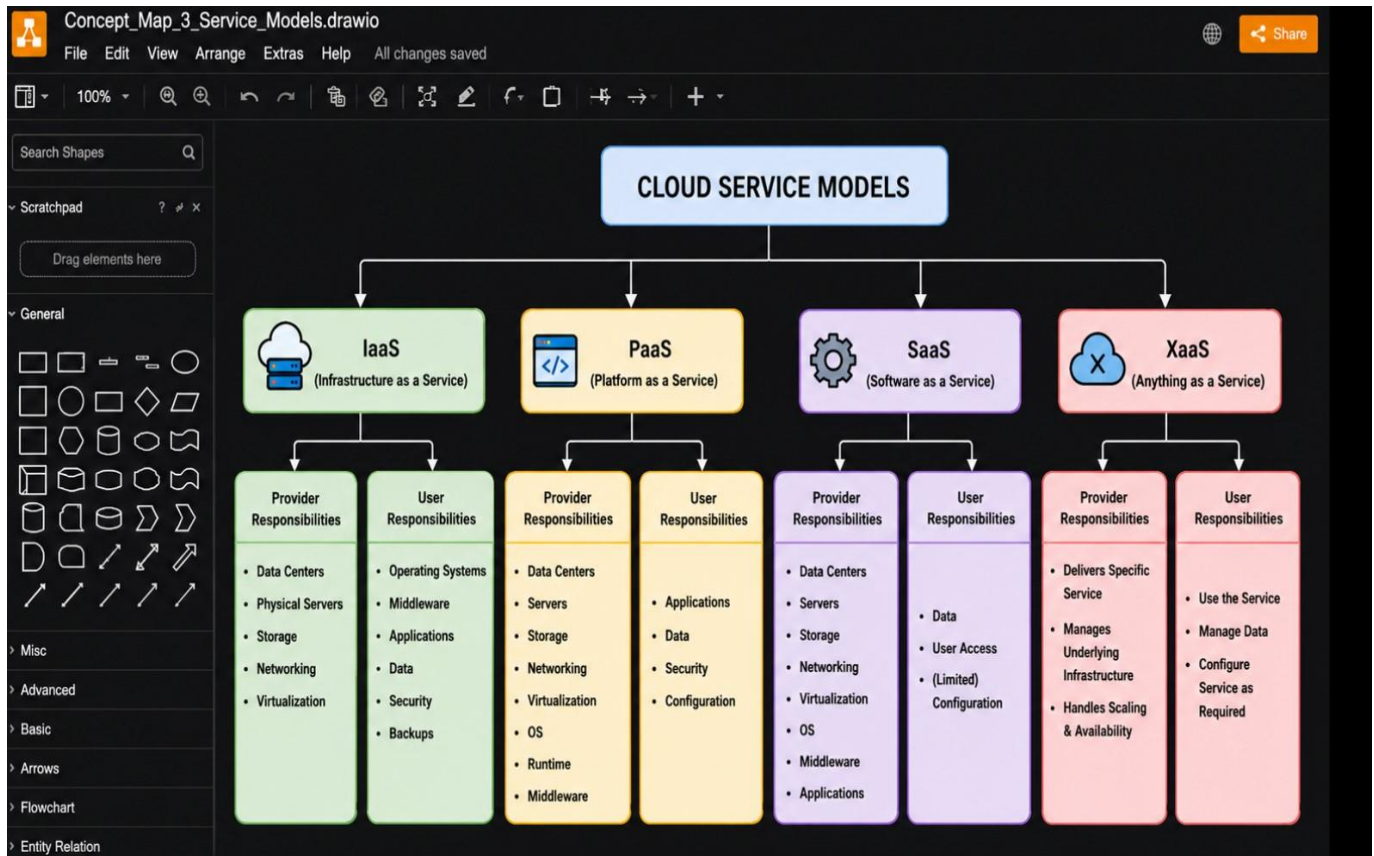
QUESTION 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations



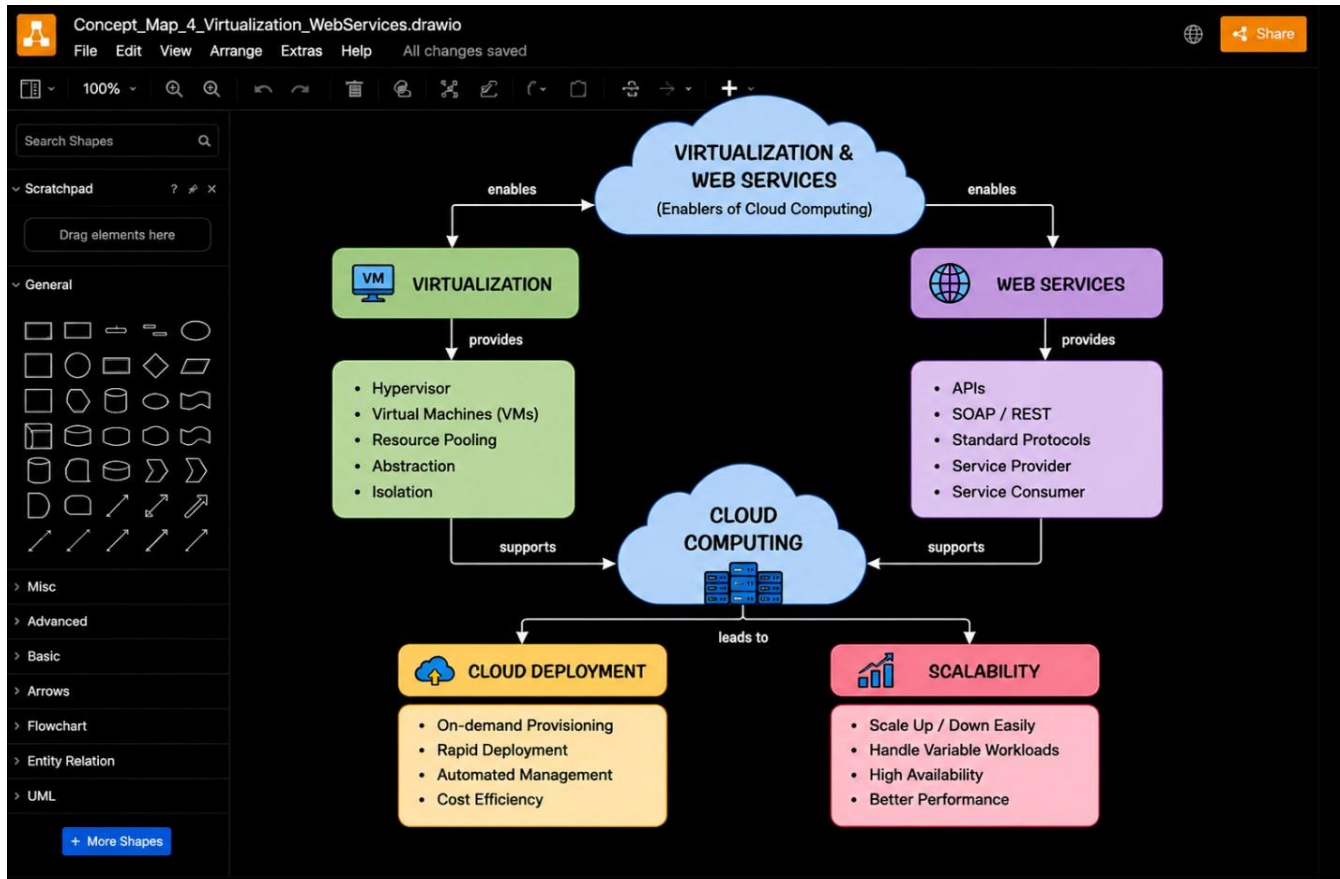
Question 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.



Question 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.



Question 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.



QUESTIONS:

1.This arrangement makes it easy to understand how each component contributes to the overall cloud architecture.

This arrangement makes it easy to understand how each component contributes to the overall cloud architecture. The design decisions that most strongly support the solution are the hierarchical layout, clear link labels, and logical grouping of related concepts. These choices improve readability, reduce complexity, and help users quickly understand the relationships between different cloud technologies. Overall, the concept map provides a complete overview of cloud computing and demonstrates how different technologies work together to deliver reliable and efficient cloud services.

2.Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

Scalability, elasticity, and resource management are essential features of cloud computing architecture. Scalability allows the system to increase or decrease computing resources to handle growing workloads without affecting performance. Elasticity automatically adjusts resources based on real-time demand, ensuring that users receive the required performance while minimizing unnecessary costs. Resource management efficiently allocates CPU, memory, storage, and network resources among multiple users and applications.

If these capabilities were absent, the cloud architecture would face several challenges. During periods of high demand, applications could become slow or unavailable because sufficient resources would not be available. During low demand, unused resources would increase operational costs. Poor resource management could lead to inefficient utilization, reduced system performance, service interruptions, and a poor user experience. Therefore, these features are critical for maintaining flexibility, reliability, and cost efficiency in cloud environments.

3. Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

The concept map includes REST APIs, SOAP web services, and cloud service models such as SaaS, PaaS, and IaaS because they represent the core technologies used in modern cloud computing. REST APIs were selected because they are lightweight, easy to integrate, and widely used for communication between web applications and cloud platforms. SOAP web services were included because they provide secure, standardized communication for enterprise-level applications that require high reliability.

The cloud service models provide different levels of functionality. SaaS delivers ready-to-use software applications, PaaS provides a platform for application development and deployment, and IaaS offers virtualized computing resources such as servers, storage, and networking. Together, these services improve system flexibility, reduce infrastructure costs, simplify application development, and enable seamless communication between different cloud components. Their integration increases the efficiency, scalability, and overall performance of cloud-based systems.

4. Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

The components in the concept map are interconnected to show how cloud computing functions as a complete ecosystem. Virtualization enables multiple virtual machines to share physical hardware efficiently, improving resource utilization. Cloud service models depend on virtualization to provide flexible computing resources to users. APIs and web services allow different cloud applications and services to exchange data securely and efficiently. Scalability and elasticity ensure that resources automatically adjust according to workload demands.

These interactions significantly improve system reliability by reducing downtime through resource redundancy and fault tolerance. Performance is enhanced because workloads are distributed efficiently across available resources. Users experience faster response times, better availability, and uninterrupted access to cloud applications even during periods of high demand. The strong relationships between these components create a cloud environment that is reliable, efficient, secure, and user-friendly.

5. Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Developing the concept map helped me understand cloud computing architecture in a more structured and practical way. Instead of learning individual topics separately, I was able to visualize how cloud actors, deployment models, service models, virtualization, APIs, web services, scalability, and elasticity are interconnected. This process improved my understanding of how cloud services are designed, managed, and delivered to users in real-world environments.

Analysing the concept map also highlighted the importance of efficient resource management, automation, and communication between cloud components. If I were redesigning the concept map, I would include more real-world cloud platforms such as AWS, Microsoft Azure, and Google Cloud Platform. I would also add cloud security concepts such as identity and access management, encryption, backup, disaster recovery, and monitoring services. Including these additional components would make the concept map more comprehensive, practical, and better suited for understanding modern cloud computing architectures.